# 41. UNILATERAL OR BILATERAL GYM STRENGTH TRAINING EXERCISES? WHICH IS BETTER TO IMPROVE CYCLING PERFORMANCE?

knowledgeiswatt.substack.com/p/41-unilateral-or-bilateral-gym-strength

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In the previous KisW Research Notes we saw that heavy gym strength training benefits cycling performance (durability included). In those studies both bilateral and unilateral leg exercises were performed. The principle of training specificity states:

'Closer the training routine to the desired performance outcome, better the results'.

Following this principle, unilateral (one leg at a time) strength training should be superior compared to bilateral (two legged) strength training, as the former mimics more the pedalling movement (pushing with one at a time).

***What is the optimal approach to gym strength training to enhance cycling performance? Unilateral or bilateral exercises?***

A study published by Ji and colleagues (German Sport University Cologne) in 2022 on *Journal of Strength and Conditioning Research* can help to answer this question. (1)

## WHAT DID THEY DO?

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- They recruited 24 trained amateur triathletes and cyclists (average age **~**30 and average VO2max **~**60)
- For a 10-week period, they were divided into three groups which performed a similar training program on the bike (6 hours per week). The only between groups difference consisted in the gym strength training they did:
    - unilateral strength group: 4x4-10 RM alternating left and right leg

      (2 sessions x week)
    - bilateral strength group: 4x4-10 RM performed the leg exercises bilaterally

      (2 sessions x week)
    - control group: no gym strength training was performed.

*RM means "repetitions maximum". 10RM - for example - means using a weight that permits to do maximum 10 repetitions.

The periodization was starting from more repetitions and less weight (10 RM) and, then, gradually progressing through the weeks to less repetition and more weight (4RM).

The gym exercise performed by both the unilateral and bilateral strength groups were the following: leg-press, leg extension, and leg curl. I am not a fan of leg-extension and leg-curl for cyclists for reasons I will specify in the comment section below, but the study can be still valid to answer to the bilateral vs unilateral question.

## WHAT DID THEY FIND?

**After the 10 training weeks, cycling performance tended to improve more in the unilateral strength (US) group compared to both the bilateral strength (BS) and control (CON) groups:**

- **Time to task failure** (read how long you can keep the power output) **at 105% of power output at second threshold** (second threshold was measured before the 10-weeks intervention):
  - US: +67%
  - BS: +43%
  - CON: +37%
- **15s sprint mean power output:**
  - US: +20%
  - BS: Unchanged
  - CON: -6%

## PRACTICAL TAKEAWAYS

**Unilateral gym strength training seems to be more effective than bilateral strength training to improve cycling performance.**

## GABRIELE'S COMMENT

There is no much more to add. These data suggest that incorporating unilateral gym strength training exercises for the legs can be a better solution compared to two-legged exercise to improve cycling performance. This is probably because unilateral training mimics more the cycling movement pattern: we produce force with one leg at a time on the bike. Therefore I recommend this kind of training in the gym.

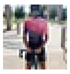
As anticipated above, personally, among the exercises used in this study I would recommend leg press but not leg extension and leg curl because they are not so specific to pedalling. Exercises like squat, lunge and leg press are more similar to the push phase on the bike.

Thanks for reading! Share with your team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

## REFERENCES:

1. Ji S, Donath L, Wahl P. Effects of Alternating Unilateral vs. Bilateral Resistance Training on Sprint and Endurance Cycling Performance in Trained Endurance Athletes: A 3-Armed, Randomized, Controlled, Pilot Trial. J Strength Cond Res. 2022 Dec 1;36(12):3280-3289.



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